

THEME: GENERAL AGRICULTURE

1. (a) Explain any four socio-economic importance of wildlife.

[4]

Question Hints and Tips

This question requires description in more detail. The response should demonstrate in more detail knowledge on the importance of wildlife in society as well as financial benefits. In this context, wildlife refers to both flora and fauna. To obtain full marks, full explanation is required for each point stated.

Model Answer:-

Aesthetic value;

Foreign currency generation;

Creates jobs,

Food/fruits/meat/honey; provides medicines, raw materials/skins/ivory/timber

Provides manure

Income generation/cash;

Recreation/sport/hunting/hospitality industries.

- (b) Describe ways of dealing with problem animals in Zimbabwe.

[3]

Question Hints and Tips

This question requires a detailed account of how animals can be dealt with. Problem animals are those wild animals which come into competition and cause serious damage to crops and livestock e.g. baboons, monkeys, quela birds but they do not pose direct harm to human life. A distinction should be made between problem animals and dangerous animals.

Model Answer:

The following are the socio-economic importance of wildlife:-

Provision of game reserves/protected areas;

Fencing to keep animals away;

Harvesting and killing; use of scarecrows;

Translocation;

Security guards/raising alarm

(c) Define the following terms as they are used in Agriculture:-

- i. Biodiversity
- iii. Ecology

- [1] ii. Poaching

[1]

Question Hints and Tips

Knowledge of terms in wildlife management is required. Concise meaning of words must be given. **Model answer:**

- i. Biodiversity

Biodiversity is the existence of many different plants and animals in an environment.

- ii. Poaching is the illegal exploitation of wildlife without a permit.

- iii. Ecology – the study of living and non-living organisms in relation to their environment.

QNI/NOV/20/4001/2

2. (a) State four importance of land as a natural heritage.

Question Hints and Tips

The question requires recall of facts on the importance of land as a natural heritage. Few words are required, with a little or no supporting arguments. Land is a factor of production that is fixed in supply.

Heritage means property that can be passed on from one generation to the next e.g. land. Property under heritage can either be natural or artificial. When these properties are passed from one generation to another at country level, it is known as national heritage.

To answer this part of the question, knowledge of the national heritage is required.

To score maximum marks you are expected to write equal or more points in relation to the mark allocation.

Model Answer -

The following are the importance of land as a natural heritage:- Crop production;

Animal production;

Shelter/ habitat;

Food;

Identify;

Unifies;

Value sites/ Great Zimbabwe as a source of national income; ensure long term utility of the resource;

Income generation for locals;

Production of curios/ souvenirs;

Wildlife; Minerals

(b) Discuss the steps followed when treating timber.

[4]

Question Hints and Tips:-

This question requires explanation of the procedures followed in timber treatment in their correct order. Give steps for either harvesting or marketing. At least 4 steps must be presented in their correct order.

Model Answer:-

Below are the steps to be followed when treating timber:-

Harvest/cut;

Debarking;

Dying/seasoning;

Use of creosote/tar/charring/partial burning;

Remove from creosote;

Cooling;

Drying

(c) Distinguish between indigenous hardwood and exotic softwood.

[4]

Question Hints and Tips

This question requires you to show differences between local hardwood timber species and exotic softwood timber species from outside Zimbabwe.

Comparison must be done on the same concept e.g. Height, colour, or weight. Hence in this case comparison can be on original strength. A table may be used. A correct and complete comparison carries one mark.

Expected response:-

Indigenous Hardwood	Exotic softwood
Originate from Zimbabwe	Originate from outside Zimbabwe
Wood is heavier	Wood is lighter
Take long to mature	Mature faster
Can survive under low rainfall	Require high rainfall
Wood is stronger	Wood is weaker

QNL/NOV/19/4001/2

3. (a) State **four** importance of the 3rd Chimurenga in Zimbabwe. [4]

Question tips and hints

- This question require a recall the 3rd Chimurenga and its application to agriculture.
- The 3rd |Chimurenga is Zimbabwe's fast track land reform programme.
- Importance in this context refers to results/ consequences; relevance worthy of note.
- You need to know what the results of this programme were. Special attention being paid to the benefits of the programme to the people of Zimbabwe. **Model Answer:-**

The importance of the third Chimurenga in Zimbabwe were as follows:-

- Relief of pressure on overpopulated land;
- Increase in agriculture production among the rural population;
- To raise standard of living in the farmer;
- To address land imbalance;

- To create employment
- To utilise unused land /idle land.

(b) Give four measures for preventing veld fires.

[4]

Question hints and tips

In this question part of the question you need to know measures for preventing veld fires. also known as bush/wild fires, have the potential to spread out of control, and cause severe damage to our environment, that is forest, thereby affecting both human and animal life. Measures are those steps taken to prevent veld fires.

Model Answer:-

Measures for preventing veld fires are:-

- Making fire guards;
- Legislation on offenders/ heavy fines;
- Education/awareness programmes;
- Establishing CAMPFIRE;
- Patrols by locals EMA/fire towers/fire alarms and - Employment of people.

(c) Describe how a planting board is used in establishing a plantation.

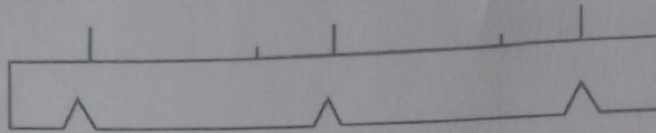
[2]

Question Hints and Tips

For this part of the question, you need a detailed account in the use of a planting board in tree planting and management. It is a tool that can help to get the proper spacing when planting tree seedlings.

To answer this question, knowledge on the use of planting board is required. The following are important points to consider:-

- Place the planting board on flat ground
- The centre notch should fit for around a peg; that represents the position of the tree;
- Drive pegs into the ground at each notch, at each of planting board; remove the centre peg: when making a planting hole;
- At planting, place the planting board over the hole;
- Place each stem of seedling tree in the centre notch and plant.



- peg three position; - put outer pegs;
- use planting board;
- remove middle peg;
- make planting hole; - bring back the planting board; - align tree wing middle peg.

Model Answer:-

How to use a planting board:-

- Peg a tree position;
- Put outer pegs;
- Mark a planting hole
- Bring back the planting board
- Align tree with middle peg
- Mark planting hole
- Bring back the planting board
- Align tree with middle peg

QN1/NOV/21/4001/2

THEME : SOIL AND WATER

4. (a) List any two sources of irrigation water

[2]

Question Hints and Tips

This question requires knowledge on sources of water irrigation. A source of water is a place from which water for irrigation can be obtained. Irrigation water – is water that is used to apply plants artificially to supplement rainfall.

Model Answer:

The following are sources of irrigation water:-

Sources of irrigation water include the following:-

Rivers/stream; Lake/dam; Weir, Well, Spring; Borehole; Old mine shaft.

- (b) Identify any **two** ways of improving soil structure.

[2]

Question Hints and Tips

This question requires knowledge on establishing/ indicating/ picking out ways of improving soil structure.

From a given list only two ways are required;

Soil structure is the arrangement of soil particles (sand, silt, and clay) to form aggregates or peds.

Structure has a direct influence on soil's physical properties which directly affect crop production for example drainage, aeration etc.

Answers:-

Ways of improving soil structure include:-

Adding manure/ adding lime; anthill soil; green manuring, crop rotation, conservation and tillage.

Model Answer:-

Ways of improving soil structure include:-

Adding manure;

Adding lime;

Anthill soil;

Green manuring;

Crop rotation;

Conservation and tillage.

- (c) Explain the importance of soil water.

[3]

Questions Hints and Tips

For this part (c) of the question, you need to know the roles of the different types of soil water (capillary and gravitational/free hygroscopic). Soil water is the forms in which water exist in the soil.

Model Answer:-

The following are the importance of soil water;

Dissolves nutrients;
Aids germination;
Cooling the plant
Regulates temperature;
Aids weathering;
Keeps plant cells turgid;
Improved microbiological
activity; Transportation of
nutrients; Helps in decomposition.

- (d) Distinguish between gravitational and capillary water.

[3]

Question Hints and Tips

For this question (a) you need to give clear cut differences between gravitational and capillary water. Gravitational is excess water either after a heavy rainfall or irrigation that move downwards due to gravitational pull. The level at which this water forms is known as the water table. Capillary water is water that moves in any direction in the soil in tubes/pores called capillary pores.

Model Answer:-

Gravitational water:-

Excess water in the soil;
Drown downwards;
By force of gravity;
Leaches nutrients Not available to
plants Capillary water:

Moves upwards;
Against force of gravity;
Held in small pores
Moves from wet to dry areas
Available to plants

Q2/NOV/19/4001/2

5. (a) With the aid of a diagram show the recommended dimensions of a contour ridge.

[4]

Question Hints and Tips

This question requires knowledge on the dimensions of a contour ridge. Dimensions are specifications and or measurements of a contour ridge such as length, breadth, and depth. The question requires the learner to draw a diagram of contour ridge and on that diagram show the different dimensions. Contour ridges are small earth structures constructed across the slope on cultivated lands.

Model Answer:-

The following are dimensions of a contour ridge:-

Buffer width 0.3m;

Height 0.25m

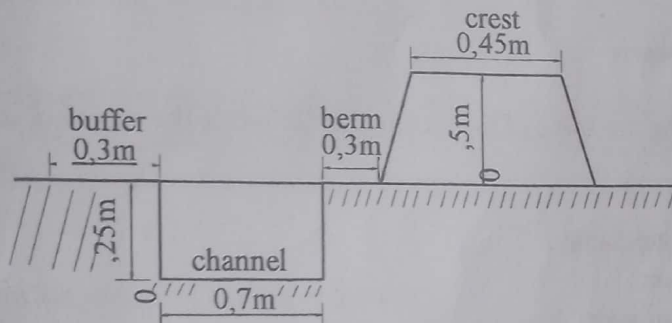
Channel width 1.7m

Berm 0.3m

Crest 0.45m

Height 0.5m

Diagram with measurements of a contour ridge.



(b) Explain how a contour ridge reduces soil erosion in arable land.

[3]

Question Hints and Tips

This question requires knowledge on how a contour ridge reduces soil erosion. To reduce is to reduce or lower the rate at which something happens. Arable land is land that is capable of growing crops. Soil erosion is a process where the upper most part of the soil is removed either by running water or wind. The following are ways in which contour ridges reduce soil erosion in arable lands:-

Model Answer:-

- Reduces speed of run off;
- Collects surface water;
- Deposits water safely away from field;
- Holds water,
- Prevents overflow downstream.

- (c) State any **three** factors that affect the choice of an irrigation scheme

[3]

Question Hints and Tips

Choice is making a decision from different available alternatives. An irrigation system is an interconnecting network of supplying water to crops e.g. Overhead and surface irrigation systems. The question requires learners to come up with factors that influence the choice/selection of an irrigation method.

Model Answer:-

The following are some of the factors that influence the choice of an irrigation system:-

- Slope/Topography;
- Size of land;
- Amount of water
- Type of soil;
- Capital/cost of material;
- Knowledge/skill; Type of crop.

Q3/NOV/19/4001/2

6. (a) Describe the application of lime during seedbed preparation.

[4]

Question hints and Tips

Learners required to give a detailed account on how lime is applied during preparation of seedbed. This is a low order question where finer details on the process of liming should be given.

Lime is a material containing calcium used in the correction of soil pH .

In order to answer this question, you need to have knowledge on liming. A detailed description on the procedure should be given. The points should be equal to or more than the marks allocated in order to obtain full marks.

Model Answer:-

Dig plough the land;

Apply lime using vicon spreader/hand

Broadcasting;

Apply in moist soil; discing into soil;

- (b) An experiment was carried out to determine the amount of air in a given soil sample. Fifty millimetres of water was added to 50mls of a sample of oven dry soil. After 30 minutes the reading in the cylinder with a mixture of water and soil was 80ml.

Calculate the percentage of air in the soil.

Question Hints and Tips.

This question requires you to use knowledge on soil constituents and how they are related. Mathematical skills on mass and volume should be applied.

To obtain full marks you are required to apply mathematical, skills and relate to the soil constituents studied, i.e. relationship between soil water and soil air. Correct units for this question should be given in the answer to obtain full marks. All working should be shown.

Model Answer:-

Soil + water = 100ml

Volume of Air = $100 - 80$
 $= 20\text{ml}$

Volume of soil = 50ml

- (c) Distinguish between field capacity and hygroscopic moistures. [4]

Question Hints and Tips

This question requires you to describe the terms field capacity and hygroscopic moistures.

Field Capacity – the amount of soil moisture or water content held in the soil after excess water has drained away, the downward movement has increased and this water is available for plant use whereas

Hygroscopic moisture is the moisture held firmly as a film on soil particles not responding to capillary action and its moisture is not available for plant use.

- (d) Explain two causes of soil erosion. [2]

Question Hints and Tips

This question requires description on the causes of washing away of top soil in land used for crop production. Knowledge on causes of erosion in different areas of agriculture is key. i.e what causes erosion in arable land can be different from what causes erosion in grazing lands.

To obtain full marks for this question, a detailed description on the erosion causes, should be given. Special attention should be paid to the mark allocation in relation to quantity of causes to be written.

Model answer:-

Removal of crop residues after harvesting;
Ploughing down the slope;
Use of heavy machinery;
Poor crop cover;
Broken conservation structures.
Broken conservation structures.

QN2/NOV/20/4001/2

7. (a) Identify any **two** ways of improving soil structure. [2]

Question Hints and Tips

This question requires knowledge on establishing/ indicating/ picking out ways of improving soil structure.

From a given list only two ways are required;

Soil structure is the arrangement of soil particles (sand, silt, and clay) to form aggregates or peds.

Structure has a direct influence on soil's physical properties which directly affect crop production for example drainage, aeration etc.

What the learner should know:

Definition of soil structure;

Types of soil structures

Improvement and maintenance of soil structure

Model Answer: -

Adding manure;
Adding lime;
anthill soil;
green manuring,
crop rotation,
Conservation and tillage.

23/01

[2]

(b) Describe any two ways of water harvesting.

Question Hints and Tips

This question requires a detailed explanation of ideas or concepts. In relation to water harvesting.

Water harvesting is the collection and storage of rain water rather than allowing it to run off.

What learners should know:

When a question says any, it implies that there are a number of possible answers.
Harvesting rain water using various methods.
Maintaining water harvesting structures
Identify ground water sources.

Model Answers:

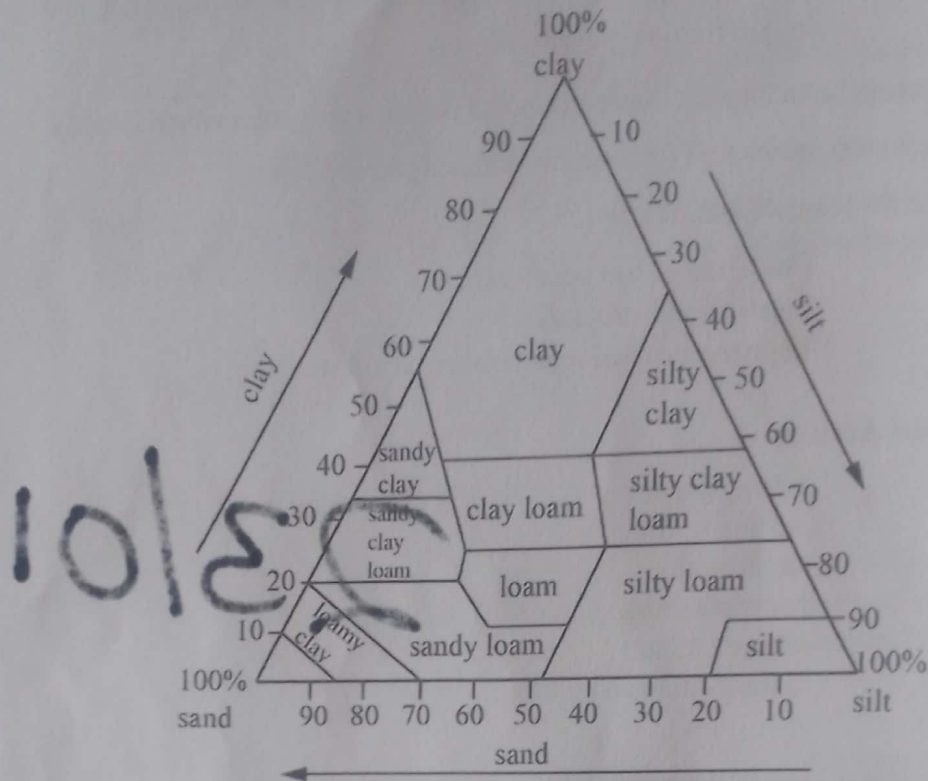
Ways of water harvesting

Construction of dams; Wells/ dhalas/domes; roof gutters.

- (c) Use a textural triangle to determine the textural class of the soils A, B and C.

[3]

[Refer to Diagram below]



Question Hints and Tips

This question requires interpretation and drawing connections among others. The candidate to use diagrams to identify the textural class of the soil.

Soil Texture refers to the fineness or coarseness of soil particles. It is the percentage of composition of sand, silt and clay in any given soil. The percentage of various soil separates should add up to 100%. Learners should know the 8 textural classes and be able to determine the textural class of a soil using the textural triangle.

Model Answer:-

A – Sand, Clay, Loam
B – Sand Clay

THEME: CROP HUSBANDRY

8. (a) State any four advantages of asexual propagation

[4]

Question Hints and Tips

Asexual propagation is the production of new plants using vegetative parts e.g. leaves, stems, cuttings, and stolons. The question requires the learner to state the benefits of using vegetative parts when reproducing new plants. It can be done using various methods including suckering, air layering, budding etc.

Model Answer:-

The following are the benefits of using asexual propagation:-

The offspring is identical to the parent/clone;

The plant established is fast.

High survival rate;

Use food from parents 'stock e.g. cuttings;

Enables propagation of sterile plants e.g. bananas;

Early maturity.

- (b) Discuss any three benefits of weeds .

[4]

Question Hints and Tips

A weed is a crop which grows where it is not wanted. You are required to give a brief description of the positive effects of weeds.

Model Answer:-

The benefits of weeds include:-

Addition of Manure/organic matter;

Medicinal purposes;

Prevent soil erosion;

Relish for humans;

Carbon sink;

Nitrogen fixation;

Mulch,

Repellent.

(c) Assess the effectiveness of biological pest control

[2]

Question Hints and Tips

The learner is expected to know the biological pest control measures. Biological pest control is the use of living organisms in the control of pests. A pest is any insect that is harmful to crops. Other pest control measures include chemical and cultural methods. Please note that both positive and negative aspects of biological control effectiveness should be given.

Model Answer

It is a slow method because predator needs to multiply faster than prey;
It is difficult to identify the correct predator
Environmentally friendly;
It is cheaper in the long run.

QN4/NOV/19/4002/2

9. (a) (i) State any two functions of plant roots. [2]
(ii) Describe the following terms as they are used in respiration. [2]
(i) Aerobic Respiration
(ii) Anaerobic Respiration
(b) Explain three importance of crop breeding. [3]
(c) Describe three advantages of haptotropism. [3]

Question Hints and Tips

This question requires that you simply recall the use of roots without further explanation. A concise answer is required. For you to be able to answer this question knowledge on functions of plant parts like roots, stem, and leaf etc. is required. The functions should be limited to the plants. The functions should be limited to the plants as functions of the roots can also be noticed in the environment.

Model Answer:

- Anchorage/ physical support
- Food storage/ storage of manufactured food.
- Absorption of dissolved minerals/ uptake of plant nutrients.

- (ii) Describe the following terms as used in respiration: [2]
i. Aerobic
ii. Anaerobic

Question Hints and Tips

This question which requires the explanation of concepts. Knowledge of respiration is required. To answer you are required to stick to respiration in plants only.

Model Answer

i. Aerobic respiration

Takes place in both cytoplasm and mitochondria of the cell;

Occurs when oxygen is available;

Occurs in all living organisms;
Source of energy for growth. ii.

Anaerobic Respiration

- No use of oxygen to break down food
- Little energy is released;
- Takes place only in the cytoplasm of the cell; - To ferment sugar to alcohol.



(b) Explain any **three** importance of crop breeding.

[3]

Question Hints and Tips

This question requires a brief description on the importance of crop improvement. Crop breeding is the science of changing traits of plants in order to produce the desired characteristics. To successfully answer this question knowledge of crop breeding is key. You should know that the proper term is tolerance not resistance e.g. drought tolerance, disease tolerance.

Model Answer:

Drought tolerant;

Disease tolerant;
Pest tolerant;
Early maturing;
Have long shelf life/lifespan;
Palatability;
High yielders;
Improved nutritional value;
Improved generic diversity.

(c) Describe three advantages of haptotropism.

[3]

This question requires knowledge and understanding of the benefits of plants' response to external environment for example touch, gravity, water and light. Knowledge of plants' response to various stimuli will help in answering the question. Understanding of the plants involved in haptotropism will help denoting the importance.

Model Answer;

Advantages of Haptotropism:-

Support their stems

Plant hold around for balance/weaker stems get balance from stronger plants;

Climbing plants grow upward to get light.

10. (a) Differentiate between open and controlled pollination
[2]

Question hints and Tips

- This question required the distinction between the open and controlled pollination.
- Knowledge on various ways of crop breeding is key. Pollination refers to the transfer of pollen grains. Differences can be tabulated for easy comparison. To get full marks a proper matching comparison should be given. Ensure you looked at, at least equal or more in relation to mark allocation.

Model Answer :- Open pollination:

- Transfer of pollen is left to nature/without man's interference.
- Self-pollination promotes undesired pollination.

Controlled pollination

- Done by man.
- Prevents undesired pollination.

- (b) Define the term heterosis

[2]

Question Tips and Hints.

This question simply requires recall of the meaning of the term heterosis. Knowledge of crop breeding terms is required. To answer this fully you are required to have knowledge on hybrid seed production as well as the benefits of hybrid seeds.

Model Answer:-

Superiority in performance of a hybrid/ offspring over the average of two parents;

Hybrid vigour.

- (c) Define the following terms

i. Emergence

[1] ii. Germination

[1]

Question Hints and Tips

This Question requires simple recall of the meanings for the given terms.

Model answer:-

i. Emergence

This is the appearance of plumule/shoot above the ground surface.

- a. **Germination** is the development of a seed into a young plant.

Q3/NOV/20/4001/2

THEME : ANIMAL HUSBANDRY

11. (a) State the part where the following processes take place in the female Reproductive system of poultry:

- | | | |
|-------|---------------------------------------|-----|
| (i) | Fertilisation | [1] |
| (ii) | Formation of chalaza | [1] |
| (iii) | Formation of inner and outer membrane | [1] |

Question Hints and Tips.

This question requires recalling of factual knowledge of the female reproductive system of poultry (Chickens). The reproductive system is made up of parts or organs involved in the production of new individuals.

To answer this question, knowledge and understanding of the reproductive parts of a hen i.e. (Ovary, infundibulum/oviduct, funnel, magnum, isthmus, shell gland, and cloaca) is required. A mark is awarded for a correctly named organ against each process.

Model Answer:-

- i) Fertilisation - Funnel/infundibulum. ii) Formation of chalaza - Magnum iii) Formation of inner and outer membrane - Isthmus

- (b) For a **named** ruminant animal, suggest classes of feeds which can be used to maintain body mass. [3]

Question Hints and Tips.

The learner should have general knowledge of ruminant animals and types/classes of feed. This is a lower order question. Class of feeds refers to the grouping of feed

materials provided to animals. Ruminants are animals having a stomach divided in four compartments.

To score all marks correct naming of a ruminant animal is required together with at least two classes of feeds.

Model Answer

Name of ruminant animal: cattle/sheep/goats.

Feeds : roughages; straight feeds; concentrates, succulents.

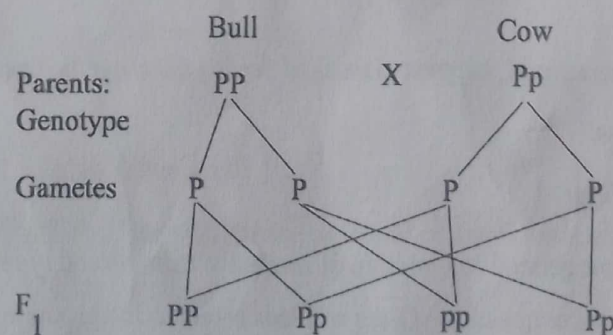
- (c) If a **PP** polled bull, mates with a **Pp** polled cow, what is the possible **F₁** generation from this cross? Show all your working. [4]

Question Hints and Tips.

In order to answer this question learners should be able to apply their knowledge to solve problems. Letters of the alphabet are used present the genes. Capital letter for dominance and small letter for recessive gene. Chosen letter of the alphabet represent the same characteristics in different ways. e.g. P – Polled, p – horned. Mating is crossing two parental genotypes. **F₁** generation is the first offspring that we get after crossing two parental genotypes.

Question requires knowledge and understanding of genetics and drawing of test crosses to come up with the desired filial generation. (**F₁**) generation are the first offspring that we get after crossing two potential genotypes. A mark will be awarded for correct crossing of parental genotypes; illustration of gametes; correct combinations to restore the diploid number of chromosomes and giving the correct genotype in the **F₁** generation.

Model Answer:-



12. (a) Identify any five uses of livestock.

[5]

This question requires the learner to list the importance /benefits of animals that are kept at the farm for example cattle, sheep, goats, donkeys, pigs and poultry.

Model Answer;

Symbol of wealth;

Source of food/milk/meat/eggs/blood;

Source of income/cash/barter trade;

Provide wool/mohair

Provide hides/skin/feathers/pelts

Social purposes/paying lobola;

Spiritual purposes/appeasing spirits;

Sources of manure;

Provide transport

Foreign currency

For exhibition/shows

(b) Discuss any three reasons for castrating livestock.

[3]

Question Hints and Tips

This question requires the learner to give benefits of rendering a male animal infertile.

Points must be clearly explained.

Model Answer:-

Animal becomes docile;

Can be trained as draught animals;

To control breeding/ avoid random mating;

Increased growth;

Improves meat/ carcass quality;

Eradicate unwanted characteristics.

(c) Give the functions of the following female reproductive organs

ii. Vagina

[1]

iii. Uterus

[1]

Question Hints and Tips

This is a low order question requiring simple listing of the uses of the given parts of the reproductive system of a cow.

You are restricted to give functions in relation to reproduction only.

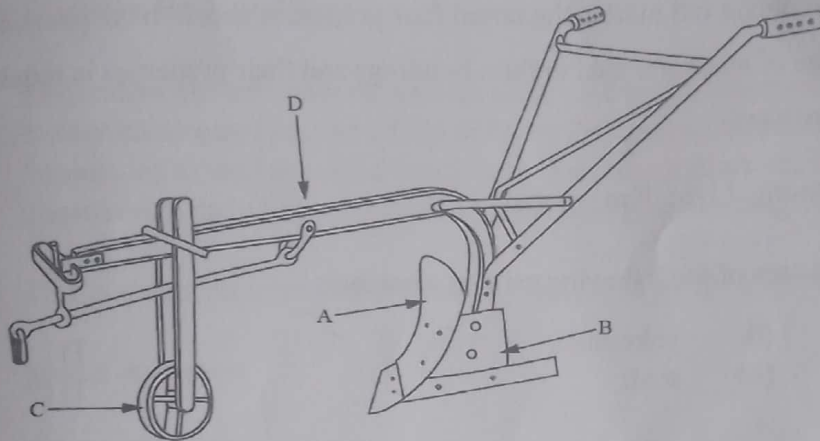
Model Answer:-

- i. Vagina – organ for insemination/accommodates the penis/copulation, Birth canal passage; ii. Uterus – region of implantation; nourishment of embryo during gestation; accommodates and carries foetus; passage of semen; relax/contract during birth; protects the embryo; produces progesterone hormone

QN4/NOV/20/4001/2

THEME : FARM STRUCTURES AND MACHINERY

13. (a) Fig. 5 shows parts of a mouldboard plough. Identify parts labelled (A-D) [4]



Identify the parts labelled A-D

Question Hints and Tips.

In order to answer this question the learner should be able to name and recall of the parts of a mouldboard plough. A mouldboard plough is a farm implement having a curved plate to break the soil, lift it and turn it upside down.

The learner should practice drawing and labelling the parts of a mouldboard plough and study the functions of the mouldboard plough.

Model Answer:-

- A – Mouldboard
- B – Frog
- C – Wheel
- D – Beam

- (b) Explain any **four** properties of iron sheets used in roof construction of farm buildings. [4]

Question Hints and Tips.

This question requires knowledge on materials used in the construction of farm buildings. Properties refers to any traits/characteristics that can be measured such as strength; durability. Iron sheets are thin plates made with iron alternating ridges, troughs used to cover roofs of buildings.

In order to obtain full marks; the raised four properties should be explained. Knowledge of materials used in farm buildings and their properties is required.

Model Answer:-

Durable/strong, Light, Fire resistance.

- (c) State the uses of the following parts of a harness:

- | | | |
|------|-------------|-----|
| (i) | yoke shaft, | [1] |
| (ii) | shaft | [1] |

Question Hints and Tips.

The question which requires a simple statement. Harness is a set of straps and fittings by which draught animal is fastened to a cart or plough. There is also a wooden cross piece that is fastened over the neck of animal(s) and attached to the plough or cart. In order to answer this question; knowledge on double neck yoke is required

Drawing and labelling parts of yokes is of ultimate importance.

Model Answer:-

- i) links two animals;
Carries the load / bears the load
Maintains the distance between the two animals Holds parts of the harness.
- ii) Attach/link trek chain to the yoke/shaft clamp; adjust for stronger and weak animals.

Q5/NOV/21/4001/2

14. (a) Identify tools for the following fencing processes: [4]

(i) Straining wire [1]

(ii) Cutting wire [1]

Question Hints and Tips

The learner should know the various tools used for fencing process. A fence is a barrier constructed around a field or farms for the purposes including marking boundaries as well as animal restriction. Various tools are used during its construction including wire strainers and digging bars among others.

This question requires you to name the tools used for the given processes.

Model Answer:-

- (i) Wire strainer
- (ii) Pliers

(c) Outline the routine maintenance of an irrigation pump. [4]

Question Hints and Tips.

This question calls the learner to recall the routine maintenance of an irrigation pump. Routine maintenance refers to activities such as inspections of machine servicing that are carried out to keep the machine in good working condition. This avoids machine breakdowns. Irrigation Pumps are used to draw water from; lower levels to higher levels or raising it to require pressure so that it can be sprayed on the fields.

Model Answer:-

- The following are the routine maintenance for in irrigation pump;
- Tightening loose bolts/nuts;
- Greasing moving parts;
- Cleaning;
- Replacing worn out parts.

(d) Outline the routine treatment procedure for wooden fencing [4]

14. (a) Identify tools for the following fencing processes: [4]

(i) Straining wire [1]

(ii) Cutting wire [1]

Question Hints and Tips

The learner should know the various tools used for fencing process. A fence is a barrier constructed around a field or farms for the purposes including marking boundaries as well as animal restriction. Various tools are used during its construction including wire strainers and digging bars among others.

This question requires you to name the tools used for the given processes.

Model Answer:-

- (i) Wire strainer
- (ii) Pliers

(c) Outline the routine maintenance of an irrigation pump. [4]

Question Hints and Tips.

This question calls the learner to recall the routine maintenance of an irrigation pump. Routine maintenance refers to activities such as inspections of machine servicing that are carried out to keep the machine in good working condition. This avoids machine breakdowns. Irrigation Pumps are used to draw water from; lower levels to higher levels or raising it to require pressure so that it can be sprayed on the fields.

Model Answer:-

- The following are the routine maintenance for in irrigation pump;
- Tightening loose bolts/nuts;
- Greasing moving parts;
- Cleaning;
- Replacing worn out parts.

(d) Outline the routine treatment procedure for wooden fencing [4]

Question Hints and Tips

The learner is expected to know the Wood Treatment Procedure. Treatment of wooden fencing is a preservative procedure involving the introduction of certain chemicals such as creosote/tar to protect timber from hazards such as wood destroying organisms like fungi and insects. This helps the fencing poles to be durable. The question requires you to give the steps followed when treating fencing poles. **Model Answer:-**

The following are the steps involved when treating fencing poles:

Cutting the desired size;

Debark the posts

Dry poles naturally/seasoning;

Immerse in boiled creosote and allow to cool; Allow the poles to dry.

Q5/NOV/19/4001/1

15. (a) Explain the functions of the listed maize sheller parts.

[5] i. roller

ii. base plate

iii. pressing fan

- iv. threshing cylinder
- v. feel tray

Question Hints and Tips

This question requires the learner to have knowledge and understanding of the parts of a maize sheller. A sheller is a machine that separates the grain from the cob/pods. There are two types of shellers. The maize sheller and the groundnuts sheller.

In order to answer this question knowledge of the parts of a maize sheller and how they function is required.

Model answer:-

- i. roller - rollers prevent clogging/blocking of the grain outlet
- ii. base plate - separates grain from cob
- iii. pressing fan - blows an air current that cleans the grain.
- iv. threshing cylinder - separates grain from the cob. Drives the shelled cobs out of the sheller.
- v. feed tray - where cobs are placed for threshing.

(b) Explain the operational principle of a maize sheller.

[2]

Question Hints and tips

This is a low order question that requires expression of ideas or concepts on the operational principles of a maize sheller.

Operation refers to the method by which maize sheller performs its functions.

In order to answer this question, knowledge and understanding of the parts of the maize sheller. The base plate and the threshing cylinder have almost similar functions because they work hand in glove. **Model Answer:-**

- Cobs are put into a feed table/tray,
- Cobs are conveyed to a rotating threshing cylinder/drum;
- Filtered with bitter bores;
- Cobs are squeezed through limited space;
- Between bitter bars and a concrete/base plate;
- Maize grain is removed from cobs; - Grain falls through holes on concave.
- Into a grain chute/outlet.
- Air is blown by the fan to remove chaff/impurities; - Grain is collected at the chute fitted with a bag.

- (c) Calculate the rolls of barbed wire required for a **four** strand fence on a plot measuring 100m x 80m. [3]

This question requires mathematical knowledge of division perimeter.

- A strand is a stretch of wire on a barbed wire fence. If a barbed wire fence has four strands . it means it has four stretches of wire running in the same direction and from each other.
- A barbed wire is a wire with clusters of short, sharp sikes set on short intervals along it.
- To be able to answer this question knowledge of perimeter (length + width) x 2 is required. Bundles of barbed wire are obtainable in definite quantities e.g. 1 bundle or 2 bundles and not as a fraction of a bundle.

Model answer:-

perimeter $(100 \text{ m} + 80 \text{ m})2;$

$$= 180 \text{ m} \times 2$$

$$= \frac{360 \text{ m}}{700} \times 4$$

$$= 2.06 \text{ rolls}$$

$$= 3 \text{ rolls}$$

QN5/NOV/20/4001/2

THEME : AGRIBUSINESS

16. (a) Explain principles of agriculture cooperatives in Zimbabwe [4]

Question Hints and tips

This question requires a detailed description of the principles in Agriculture cooperatives. A cooperative is a group of people coming together for a common cause, e.g. economic cause. Principles are laws that serve as a common foundation for a system. Agriculture cooperatives are run, owned and democratically controlled by the members. They can be farming, transport, or savings.

Model Answer:-

Below are the principles of agriculture cooperatives:-

- Open membership;
- Equal voting rights/democracy;
- All members buy /own shares;
- Members buy/own shares;
- Members get training on management;
- Shares profits/losses;
- Religious; political neutrality.

(b) Outline the steps in decision making process in their correct sequence. [4]

Question Hints and Tips

Decision making is a process of making choices between two or more courses of action.

A step by Step decision making process can help to make more thoughtful decisions

Model Answer:-

- Identify problems;
- Gather information;
- Identify alternatives;
- Choose alternatives;
- Take action; review decision/consequences/evaluation.

(c) Explain the importance of farm budgeting [2]

Budgeting is the technique for estimating future income and expenditure. You are therefore required to give the benefits of budgeting in agriculture enterprises.

Model Answer:-

Below are the importance of farm budgeting:

Evaluation of farmer's performance;
Help in decision making and choices made/planning

Helps the farmer to secure loans; To find income and expenditure; uncover expenses that could be overlooked during production; reminds the farmer to pay bills/debts.

Q6/NOV/19/4001/2

17. (a) The output for sorghum and maize in Zimbabwe improved during the 2016/17 agricultural season. Suggest any **four** factors contributing to the increase. [4]

Question Hints and Tips

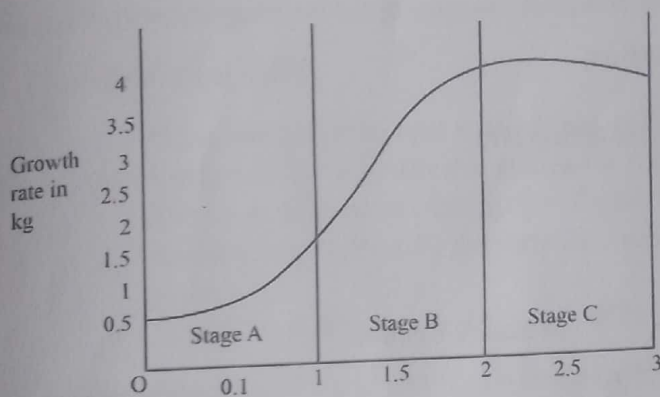
This question requires use of acquired information in the new situations.
Knowledge on the factors affecting supply of agricultural produce is required.
Improvement means increase in agricultural output/yields.
Knowledge of the following is required:-

Describe the law of supply
Interpret supply curve;
Factors affecting supply.

Model Answer:-

Availability of inputs;
Availability of irrigation;
High producer price;
Technological advancement;
Training/awareness;
Favourable weather conditions temperature/adequate rainfall;
Availability of improved varieties;
Government subsidies/ input schemes by government.

(b) Fig. 6 shows the relationship between feed consumed and growth rate of broilers.



- (i) Which stage in Fig. 6 is economically viable to sell broilers? [1]
(ii) Why is the stage in (b) (i) the most appropriate? [1]

Question Hints and Tips

It is a low order question that requires you to establish the most appropriate stage which is economically viable. Economically viable refers to a stage that is feasible and determined that it has more benefits than costs, based on the Cost benefit

Analysis. To answer this question the following is required to:-

- Explain the law of diminishing returns
- Interpret the law of diminishing returns
- Describe the implications of diminishing returns

Model answer

- i) Stage B;
 - ii) Increased application of feeds tends to maximum weight; return per input will start to diminish.
- (c) For either a **pig** or a **donkey**, design a breeding record. [4]

Question Hints and Tips

This question requires knowledge on production of new or original work. Either means only one studied animal should be addressed by underlining it on the question or writing it. A design is a plan to be followed. A record is a set of data relating to an individual animal.

To answer the question, the following knowledge of compiling physical and financial records.

Model Answers

- Name/identification;
- Age of animal
- Date of farrowing/giving birth
- Weaning date;
- Mating date;
- Size of litter;
- Birth weight;
- Name of sire; - Weaning weight.

Q6/NOV/20/4001/2

18. (a) List any **three** importance of financial records to a farmer. [3]

(b) A farmer harvested **6 tonnes** of maize and sold it at **\$300** per tonne. He then made the following payments.

Compound D	- \$200,00
Chemicals	- \$ 50,00
Farm manager's salary	- \$400,00
Urea	- \$150,00
Farm licence	- \$100,00

Calculate the gross margin for the enterprise. (Show all working) [3]

Question Hints and Tips

Knowledge of farm records and account will help to answer the question fully. This question is a low order question that requires you to state in clear terms.

Financial records shows information related to money coming in and money used at the farm.

Importance are benefits of keeping the records.

Model Answer:-

- (a) determine profit/loss;
planning/decision making;
determine
income/expenditure; secure
loans; budgeting;
- (b) This is a high order question which requires critical thinking and application of mathematical skills.

Gross margin is the total income less variable costs.

Variable costs are costs that comes for a specific enterprise.

The formula for gross margin is = **Gross Income – Variable costs** You rare required to identify the variable costs from the list of expenses given.

Fixed costs such as salaries do not affect gross margin.

$$\begin{aligned} \text{GM} &= \text{GI} - \text{TVC (Compound D, Chemicals, urea)} \\ &= (300 \times 6) - (200 + 50 + 150); \\ &= \$1\ 800 - \$400; \\ &= \$1\ 400 \end{aligned}$$

- (c) Identify any two possible causes of changes in prices of agricultural products.
[2]

Question Hints and Tips

This is a low order question which requires indicating courses of changes in prices of agricultural products.

A price is an attached value of a good/service.

A change in price is a fluctuation either increasing or decreasing. In order to answer this question knowledge on the theory of supply and demand is important; factors affecting supply and demand of agricultural products.

Model answer:

(d)

seasons
supply
shortages/gl

uts

price of substitute
demand
substitutes
demand
subsidies cost of
production availability
of substitutes
level of income
level of technology
perishability
quality of products

Q6/NOV/20/4001/2

SECTION B OPTION 1: CROP HUSBANDRY

19. (a) Outline the advantages and disadvantages of organic fertiliser in crop production.

[9]

Question Hints and Tips

This question restricts the answer to giving essential. Advantages simply means benefits derived from using fertilisers.

A disadvantage is a loss or unfavourable condition associated with the use of organic fertilisers.

Organic fertilisers are those derived from plant and animal material for example compost; cattle manure, poultry dropping.

In order to answer this question, knowledge on the following is important

Differences between organic and inorganic fertilisers.

Types of organic fertilisers.

NB the question can be answered in table form.

Model Answer:-

The following are the advantages and disadvantages of organic fertilisers in crop production:- Advantages:

readily available; relatively cheap;
contain wide range of nutrients;
improves soil structure; improves water infiltration; improves aeration/drainage; source of food for soil organisms; reduce soil erosion; regulates soil temperature; reduce leaching of nutrients;
improves water holding capacity;

Disadvantages

bulky; laborious to apply; take long to prepare; can spread diseases; can spread weeds;
supply unknown quantities of nutrients;

(b) Evaluate the use of silos in grain storage.

[5]

Question Hints and Tips

This is a high order question that requires justification of a stand or decision

An evaluation is to judge the value of or worth of something.

Silos are concrete towers on a farm used to store grain such as maize, groundnuts,

groundnuts, wheat, sorghum, soya beans, and field beans.

In order to answer this question, knowledge on storage structures for crops is required

Tip: Consider advantages and disadvantages. Answers can be presented in tabular form.

Advantages

- Large volumes of grain are stored;
- Not vulnerable to pests or rodents;
- Protection against fire;
- More secure;
- Durable;
- Damp free/not susceptible to moulds,

Disadvantages

- High capital costs;
- Difficult to construct;
- Cleaning can be challenging;
- Not flexible.

The following are ways of reducing the effects of environmental factors on crop production.

Question Hints and Tips

The question requires you to give an account on how to reduce effects of environmental factors on crop production.

Environmental factors are any biotic or abiotic factors that affect crop production for example wind, temperature, rainfall, humidity.

Reducing is however, making smaller or lessening the effects of something

Model answer:-

The following are ways of reducing effects of environmental factors:-

shading; mulching; pot
holing; manuring; tie-
ridging; watering/irrigation;
grow drought tolerant
plants; cover crops; wind
breaks; conservation tillage;
green houses; cloud seeding
creating small fires/burn grass/tyres;
use of water bottles; use of frost
barriers; raised beds/ridges open
ditches/drainage pipes

QN/9/NOV/20/4001/2

20. (a) Describe toxicity levels of Agro-chemicals.

[8]

Question hints and tips

- This question requires giving a detailed account in words and also revealing more relevant facts on toxicity levels.
- Toxicity refers to degree of being poisonous. Agro chemicals are chemicals used in agriculture such as pesticides.
- It is its capacity or ability to cause injury or illness. In order to answer this question you need to know:-
- Precautionary measures taken when using and storing agrochemicals.

Model answer;-

- Green - caution/harmful/less toxicity
- Orange/amber - poison/dangerous
- Red - very poisonous/very dangerous
- Purple - extremely poisonous/extremely dangerous

(b) Explain advantages of conservation tillage.

[7]

Model Answer:-

The following are the advantages of tillage - -

- reduces erosion;
- improve soil structure;
- reduce runoff;
- conserves moisture;
- improve drainage;
- reduces evaporation;
- reduces leaching
- improves infiltration/
- improves drainage and aeration;
- adds fertility;
- raises soil temperature/regulates temp;
- improves microbial activity;

- saves time;
- reduces operational costs;

c) Design, in sequence, a **four** year rotation for a field crop.

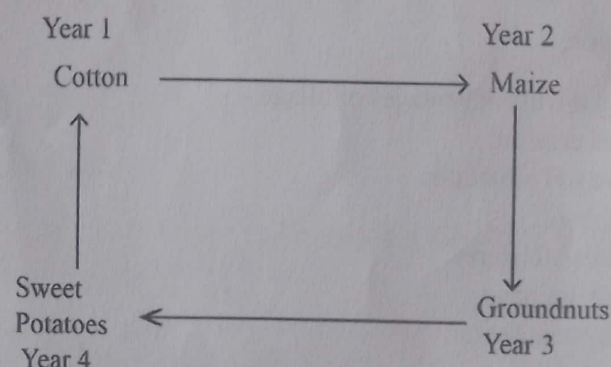
[5]

Question Hints and Tips

To answer this question you need to know that:-

Crop rotation is the practice of growing a series of different types of crops on the same piece of land in a planned sequence. You need to know the principles to be followed when designing a four year crop rotation for example crops of the same family should not follow each other. Deep rooted crops should follow shallow rooted crops. Always include a legume crop in your rotation. A sequence is illustrated by numerical figures e.g. year 1; year 2; year 3; year 4; or arrows which show direction of flow. A field crop is an agricultural crop e.g. maize grown on large areas.

Diagram (Crop Rotation)



Model Answer

cotton; maize;
groundnuts;
sweet potatoes;
correct sequence;
to include legume;
time/season/year;

21. (a) Describe any five branches of horticulture

Q8/NOV/20/4001/2

[10]

Question Hints and Tips.

The learner is required to come up with branches of horticulture. For each branch you are required to write a brief description. Horticulture is the art of cultivating plants in the garden.

Answers: the following are branches of horticulture.

Olericulture – growing of vegetables e.g. rape/covo.

Floriculture – is the growing of flowers e.g. marigold.

Pomology- is the growing of trees e.g. apples/mangoes.

Arboriculture – Ornamental growing of trees/shrubs/hedges.

Landscaping – designing land for decoration/planning/planting.

Nursery Culture – production of seedlings e.g. vegetables; fruit trees/ornaments.

Viticulture – production of vines.

- (b) Discuss why deciduous fruit trees are pruned.

[10]

Question Hints and Tips

The learner is required to provide the reasons why the practice of pruning is done on deciduous fruit trees. A deciduous fruit tree is one that sheds annually. Pruning is the act of trimming by cutting away dead or overgrown branches. e.g of apple tree.

Model Answers:-

Deciduous fruit trees are pruned in order to:

Encourage formulation of new branches

Helps maintain deciduous tree health and productive

Creates a desirable structure for supporting fruit crop

Keeps the tree more manageable

Responses vigorously when large limbs are removed

23/01

- (c) Describe the preparation of a planting hole for a named fruit tree. [10]

Question Hints and Tips

Model Answer :-

Banana/Guavas/Mango/Grape/Orange/Naartjies/apples

Marking; 60 -100cm x 100cm deep x 60cm – 100cm wide.

Separate top soil from sub soil.

Put mixture into hole;

Put subsoil and make a ridge/basin

Clearing;

Digging;

Use of a plant board.

Q7/NOV/19/4001/2

22. (a) Explain four factors considered when selecting a suitable site for an orchard. [10]

Question Hints and Tips

The learner is required to come up with factors considered when selecting a site for an orchard. These factors should be explained in detail so as to get all marks. A factor is a circumstance that contributes to a result. Site on an area of ground on which the orchard is established. Orchard is a piece of ground on which the orchard is established. Orchard is a piece of land planted with fruit trees. Each factor has to be accompanied by an explanation.

Model Answers:

Consider the following factors:

Near source of water, for irrigation;

Fertile soil; for growth; fast growth; high yield; reduced costs; slope;

to prevent erosion; water logging;

Well drained; to prevent water logging thieves; stray animals; Wind

break to prevent lodging; dropping of flowers, /fruits.

- (b) Examine the marketing strategies employed in cut flowers or fruits. [10]

Question Hints and Tips

The question requires one to come up with strategies used to market cut flowers or fruits. Marketing is the act of buying and selling and is influenced by the forces of demand and supply. A strategy is a technique used to market products. Knowledge on horticulture is important.

Model Answer-

The following are some of the marketing strategies:-

Cleaning - rinse in water; to remove dirt.

Grading - according to size/ colour/rejects/freshness; quality

Weighing - into quantity demanded by buyers

Packaging - wrapping with plastic/boxes/crate/trays/baskets

Storage - keep fresh/cool/dry/cold room/use of

hormones Transport - reliable; containers; secure;

licensed Pricing - market demand; small profit quick returns.

Advertising - electronic media/social media/posters/radios

Processing - value addition; branding/ labelling; to identify product with produce.

Q9/NOV/19/4001/2

23. (a) Outline the socio-ecological importance of forestry [8]

Question Hints and Tips

This is a higher order question that restricts the answers to giving essentials. It is a general description showing the essential features of something but not in detail.

Socio-economic, these are two terms combined (society and ecology).

Society is a group of people living together in a more or less ordered community.

Ecological refers to the relations of organisms to one another and to their physical surroundings.

Forestry is guiding term in this question; socio-ecological benefits as related to forestry.

In order to answer this question knowledge with understanding of the following is important. Socio, economic, cultural and ecological importance of forests;

Major forests in Zimbabwe are:- Indigenous and exotic forests.

Model Answer, the following are the socio-ecological importance of forestry
Fruits/food/honey
Shade;
Timber/poles;
Firewood/fuel/animal habits;
source of medicine; control
soil erosion; rain formation;
reduces toxic substances/pollution/gaseous exchange; balanced
ecosystem recycling nutrients; aesthetic value; indicates the
status of soil/fertiliser/water availability/soil depth.

(b) Discuss the process of pricking out in the gum tree and establishment.

[4]

Question Hints and Tips

This is a high order question that requires investigation by reasoning; argument and explanation.

Pricking out is the transfer of seedlings from one seedbed/selected box to another seedbed/selected box to reduce competition for water; air; sunlight; space; nutrients. (Limiting factors). It is the separating out of seedlings growing together and transferring them to pots of potting mix.

In order to answer this question knowledge with understanding on the following is important.

Tree nursery establishment and management.

Hint: A sequence of events to be followed is required.

Model Answer:

The following is the process of pricking out in gum trees.
Fill polythene bags/trays with soil;
Water polythene bags/trays;
12 hours before pricking out;
Mark centre of polythene bags/trays
Transfer/lifting seedlings
Place into a hole of polythene bag/tray;
Cover root/ from soil
Provide shade; water the polythene bags/trays.

- (d) Describe and explain the management practices carried out in a gum tree plantation. [8]

Question Hints and Tips

This question requires giving a detailed account in words and also revealing more relevant facts on the management practices carried out in the gum tree plantation. Explanation requires classification of the context (management practices) or giving reasons,
Management is the conducting or supervising of something.

In order to answer this question knowledge with understanding on the following is important:

Model answer:-

Management Practices carried out in gum tree plantations are as follows:-

- Construction of fire guards;
- To prevent wild fires;
- Fencing to prevent livestock; Pest control to control termites.
- Weeding to reduce competition (for water, light, space, nutrients)
- Pruning/branch management to shape the tree
- Gap filling to maintain plant population.
- Two shoots per stool;
- Stool coppice management;
- Spray using deltrin/aldrin
- Watering to prevent wilting/provide moisture.

Q7/NOV/20/4001/2

24. (a) Describe the characteristics of viral and fungi diseases in crop production. [10]
- (b) Outline the importance of tropisms in crop growth. [10]

Question Hints and Tips

This question requires knowledge on characteristics of plant diseases (Fungal, Bacterial and Viral). A characteristic is a distinguishing feature of a disease. In order to answer this question the learners should have comprehensive knowledge of the general symptoms of fungal, bacterial and viral diseases. The question should be answered as an essay in paragraph form.

Model Answer:-

The following should be included:-

Viral – can survive inside living tissues;

- Cause plant diseases;
- Disrupt the growth of plants/deforming and weakening;
- Multiply rapidly;
- Spread by suckling insects/aphids and hands;
- Cause ring sports;
- Cause rosette/shortening of stem;
- Curved leaves;
- Reduced yield;
- Mosaic/dicolouration; - Stunted growth.

Fungal – Spread through spores

- Prevalent in hot and humid conditions;
- Causes rots/wilting;
- Causes sports;
- Causes rust; - Lasts long in the soil; - Reduced yield.

Q7/NOV/214002/

2

(b) Importance of tropisms in crop growth

[10]

Question Hints and Tips

This is testing on the ability to give essential details in a few words. Tropisms are plant growth responses to external stimuli. A stimulus is an external phenomenon that has an influence on a psychological activity.

In order for the learner to obtain more marks, the following is expected:-

- Describe the plant tropisms;
- Demonstrate the plant properties to tropism; - Evaluate the importance of plant tropisms.

NB: A response can either be positive (towards the stimuli) or negative (away from the stimuli). Marks are awarded for identifying and then explaining the tropisms.

Model Answer:-

- Geotropism – causes roots to penetrate soils/find water/nutrients;
- Gives plants physical support;
- Negative geotropism causes shoots/germinating seeds to grow up.

Phototropisms – Need light for photosynthesis;

- Causes shoots to reach up
 - Shoots are exposed to gaseous exchange to take place;
- Pigmotropism - enables climbing plants to attach to support; stems of plants coil around a form of support.

Hydrotropism - enables roots to penetrate soils in search of water.

Q7/NOV/21/4001/2

25. (a) Discuss the importance of land preparation in crop production. [7]

Question Hints and Tips

This question requires a **debate or argument**. Importance refers to the condition or quality of being worthy. Land preparation is the processing of the land to provide a favourable environment for seed germination and growth.

The following concepts are of paramount importance:

Reasons for land preparation;
Land preparation procedures;
Tillage methods.

Expose pests to heat/cold
To destroy growing plants/ weeds; bury crop residues;
To incorporate manure/fertilisers/lime;
To level land;
To break pans/cap
To improve aeration; improve water retention;
To improve water infiltration/drainage;
To create fine tilth;
To promote easy root;
Improves soil fertility;
Improve seed and soil contact.

- (b) Assess the need for hybrid seeds in Agriculture. [5]

Question Hints and Tips.

This question requires justification of one's stand or decision made. Hybrid seed is the first generation of seed of a cross produced by controlling the pollination and by combining two or more lines, varieties or species.

The learner should have knowledge on the following:

Crop breeding;
Hybrids

Heterosis in plant
Open and controlled breeding

Population follows a geometric progression whilst food production follows an arithmetic progression.

Model answer:-

- Better yields;
- Improved market value/better profit;
- Tolerate to diseases
- Tolerance to pests;
- Grow faster than open pollinated varieties.
- Drought tolerant;
- Improved quality

- (c) Describe any **four** methods of fertiliser application.

[8]

Question Hints and Tips.

Describing requires a brief account of the methods used when applying fertilisers to crops. A fertiliser is any material added to the soil to supply nutrients in their right known proportions to plants.

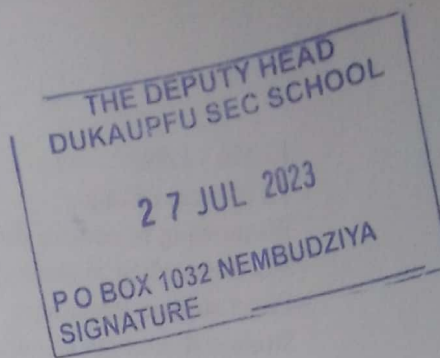
To obtain full marks the learner is required to identify the method of fertiliser application as well as the correct explanation of that method. Methods required in this question are given and bolded.

Model Answer;

The following are methods of fertiliser application:-

- Broadcasting – by scattering using hands/spreader;
- Hill placement – place a hill next to the plant;
- Banding – place fertilisers below seed during sowing in a row;
- Foliar Spray – apply fertiliser mixed with water through irrigation;
- Fetigation – apply fertiliser mixed with water through irrigation;
- Dolloping – making a hole next to the plant and apply fertiliser to it;
- Side dressing – apply fertiliser in a row next to the plant;

Q8/NOV/21/4002/1



26. (a) Describe the management of a named cereal crop from germination to maturity.

[10]

Question Hints and Tips

The question requires the learner to come up with the management practices that are followed from germination to maturity of a named Cereal Crop. Marks will be awarded for naming the crop. Management practices should be between germination and maturity. One cereal crop to be studied from the following; - Maize/Sorghum/Wheat.

Model Answer:

Name of Crop: Maize/Sorghum/Wheat.

Irrigation; named method of irrigation;
gap filling;
weed control; named method of weed control; named example; pest control; named pest; named example
topdressing: named method of application, named fertilisers;
scouting disease control; named disease, named method named records

- (b) Discuss the storage of a named legume crop after harvesting.

[10]

Question Hints and Tips

One legume crop to be studied from the syllabus (Field beans; Soyabeans, groundnuts). Naming of the Crop under study is required. Knowledge on how the Crop is stored is required. Storage is keeping for future use.

Model Answer:-

Storage of legume crops involve the following:-
Name of Legume crop: soya beans/field beans/groundnuts
Named Storage:- Moisture content;

12.5% - 12%
To prevent rotting
Winnowing to remove dirt;
Store according to grades;
Clean storage structures;
Storage should not allow moisture;
Store in bags;
Control storage pests; named example;/chemical
Fumigation/dusting/named chemical

Q8/NOV/19/4002/2

27. (a) Describe the management of a named cereal crop from germination to maturity. [10]

Question Hints and Tips

The question requires the learner to come up with the management practices that are followed from germination to maturity of a named Cereal Crop. Marks will be awarded for naming the crop. Management practices should be between germination and maturity. One cereal crop to be studied from the following; -
Maize/Sorghum/Wheat.

Model Answer:

Name of Crop: Maize/Sorghum/Wheat.

Irrigation; named method of irrigation;
gap filling;
weed control; named method of weed control; named example; pest control; named pest; named example
topdressing: named method of application, named fertilises;
scouting disease control; named disease, named method named records

(b) Discuss the storage of a named legume crop after harvesting.

[10]

Question Hints and Tips

One legume crop to be studied from the syllabus (Field beans; Soya beans, groundnuts). Naming of the Crop under study is required. Knowledge on how the Crop is stored is required. Storage is keeping for future use.

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content;
12.5% - 12%
To prevent rotting

Winnowing to remove dirt;
Store according to grades;
Clean storage structures;
Storage should not allow moisture;
Store in bags;
Control storage pests; named example;
Chemical fumigation/dusting/named chemical.

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- 28 (a) Describe the effects of the following groups of pests on crop production:
- (i) biting and chewing pests, [3]
 - (ii) piercing and sucking pests, [3]
 - (iii) boring pests. [2]

Question Hints and Tips.

This question requires a detailed account in words on the effects of pests on crops. An effect is a result whether negative or positive. The learner should have a clear understanding on the different classes of pests and their feeding habits.

Model Answer-

- (i) Biting and Chewing

Cut the young stem;
Reduce leaf quality;
Reduce leaf surface area;
Reduces yield;
Transmit diseases e.g. locusts/grasshopper/termites.

- (ii) Piercing and Sucking

damage through removal of sap;
causes leaf curl; there is little mechanical damage to the
plant tissues; saliva injected is toxic/ may transmit disease
causing microorganisms/viruses; reduced yield;
reduced quality;

- (iii) Boring pests

Make holes in the plant stem/fruit;
Cause plant lodging;
Reduced quality;
Reduced yield;
Causes rotting
Causes wilting
Example maize stock borer

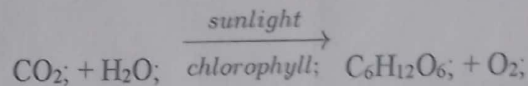
Q9/NOV/21/4001/2

- (b) (i) Give the chemical equation of photosynthesis. [5]

Question Hints and Tips

A chemical equation is the symbolic representation of a chemical reaction in the form of symbols formulae. Where in the reactant entities are given on the left hand side and the product entities are given on the right hand side with a plus sign in the entities in both the reactant products.

Possible answer:-



- (iii) Explain the roles of chlorophyll, carbon dioxide, water and light in photosynthesis. [7]

Question Hints and Tips

A simple recall of facts is required. Recalling of a balanced chemical equation is required. Explaining is making an idea clear by describing it in more detail. Ideas on chlorophyll; carbon dioxide; water and light in photosynthesis. The learner should have comprehensive knowledge on the process of photosynthesis taking note of the raw materials and by products of the process.

Model Answer:-

Chlorophyll:

Gives green colour in plants;

Absorbs energy of sunlight

Is contained in the chloroplasts

Amount of chlorophyll affects the rate of photosynthesis

Chlorophyll levels causes a reduction in net photosynthesis.

Carbon dioxide

- Combines with water;
- Is a source of carbon;
- Used to make carbohydrates
- Stabilises the chlorophyll
- Required during chemical reactions

i) Light

- Splits the water molecule/photolysis;
- Source of energy;
- An increase in sunlight intensity caused increase in rate of photosynthesis.

QN9/NOV/21/4001/2

SECTION B OPTION 2: ANIMAL HUSBANDRY

29. (a) Describe the slaughtering and dressing of either **rabbits/layers/indigenous** chickens.

[8]

Question Hints and Tips

Slaughtering is the killing of animals usually referring to killing of domestic livestock such as poultry. Dressing refers to meet the market demands. You are required to highlight the steps followed when slaughtering and dressing poultry.

the following procedure is done on slaughtering and dressing of:-

Layers/indigenous chicken

Method: stunning/distortion;

cut off head with knife;
hold the carcass with neck facing
down; allow bleeding; scald/immerse
in hot/boiled water; pluck feathers;

remove internal parts/liver, heart,
gizzard; cut off legs; clean the carcass;
fold; wrap in a plastic; refrigerate/freezer;

Rabbits

starve animal for 12 hours

put on flat structure with ears held

forward; stun the animal behind ears; cut
off head; allow bleeding;

hang the carcass neck down for

skinning; remove offals/evisceration;;

flash/wash with water; cut of wrists cut

to required pieces; weighing

grading pack
into plastics

30. (a) Discuss the importance of hygiene during the slaughtering of the animal in (a).

[9]

Question Hints and Tips

Hygiene means conditions or practices conducive to maintaining health and preventing disease; especially through cleanliness. During slaughtering hygiene has to be maintained for various reasons including preventing spread of pathogens. In this question you are expected to give the benefits of maintaining a hygienic environment during the slaughtering of the animal above.

Model Answer:-

Below are the importance of hygiene in slaughtering of poultry and rabbits.

to avoid carcass contamination which may lead to food poisoning.
To reduce losses.
Prevent spoilage/putrefaction/prevent rotting
Secure meat of good quality;
Reduces odours/keep away odours. Achieve high quality grade meat;
Lengthen shelf life;
keep away flies
equipment to be thoroughly cleaned;

- (c) A live bird weighs 2,5 kg. After dressing its carcass weighs 2.0 kg. Calculate the dressing percentage.

Question Hints and tips

A live bird is a bird before slaughter. Carcass refers to the body of a dead animal especially livestock which includes poultry. Dressing percentage is obtained by dividing the warm carcass weight by the shrunk live weight of a bird and expressing the result as a percentage. The question requires mathematical skills.

Model Answer:-

$$\frac{\text{carcass weight (dressed mass)}}{\text{live mass}} \times 100$$
$$\frac{2.0}{2.5} \times 100$$

31. (a) Describe the nutritional requirements of either a pig or donkey.
[10]

Question Hints and Tips

Nutritional requirements are the amount of each nutrient needed in the body of an animal. The question requires you to state the nutrients and give their functions in the body of an animal.

Model Answer:-

- Below are the nutrients required:
- Carbohydrates for energy;
- Protein for body building/repair of worn out tissues;
- Vitamins for maintain a healthy body;
- Minerals; e.g. calcium for bone formation;
- Fats for energy/warmth/protection of delicate organs;
- Water for digestion/coolant/part of blood and other body fluids; -
Fibre/roughage for easy digestion.

- (b) Discuss a named bacterial disease of livestock under the following sub-headings:

- | | |
|------------------------------|-----|
| (i) Name of disease, | [1] |
| (ii) causal agent, | [1] |
| (iii) signs and symptoms, | [4] |
| (iv) prevention and control. | [4] |

Question Hints and Tips

Model Answer:-

- (b) Name of disease
- (i) Anthrax;
- (ii) causal agent:
Bacillus anthracis;
- (iii) Signs and symptoms:
- sudden death/high mortality rate
 - high body temperature/fever;

- trembling;
- diarrhoea;
- blood comes from orifices of carcasses;
- rapid decomposition of carcass - no rigor mortis/stiffness of meat; - blood does not clot.

(iv) Prevention and Control

vaccination;
 report to authorities/import control/police/vet offices
 Destroy all carcasses/burn/ bury;
 quarantine;
 control use of antibiotic;
 named example;
 Use of antibiotics e.g. penicillin/procain/Hilet/Teramycin/Bennyl

32. (a) Describe characteristics that make exotic breeds of cattle suitable for Zimbabwe. [10]

Question hints and tips

Characteristic are features or qualities typically belonging to certain breeds serving as identification feature.

Exotic breeds of cattle are those breeds that originate from other countries for example Brahman. We also have indigenous breeds that are native to Africa for example Mashona and Tuli among others.

You are required to give a list of features that make exotic breeds.

Model Answer:-

Exotic features have the following characteristics:-

High birth weight;
 faster growth rates;
 mature earlier;
 high milk
 yield/meat/carcass/weight; high
 milk quality high weaning weight
 high fertility rate
 low milk/meat/carcass;
 high feed conversion ratio;
 feed conversion ratio;

low

high turnover
turnover;
mature earlier

- (b) Discuss natural and artificial acquired immunity.

[10]

Question Hints and Tips

When antibodies are passed from mother to young one during birth or in colostrum, it is referred to as passive immunity. When an animal recovers naturally from an infection it is referred to as active immunity.

Model answer:

Natural

Passive immunity; female mammal give antibodies to a young in womb; colostrum give antibodies to young animals; active immunity; animal produce antibodies after an attack by disease; lasts longer;

Artificial

Passive immunity; use of serum/blood extract from animal immunity; gives immediate protection; active immunity; use of vaccine; lasts for a long time;

Q12/NOV/19/4001/2

33. (a) Describe factors considered when siting a housing unit for a **named** ruminant.

[8]

Learners should know that:

To effectively answer the question, you are required to identify factors to consider when constructing a housing unit for a ruminant of your choice for example cattle, sheep or goats.

You are required to name the ruminant under study and then explain the listed factors.

Model answer:

Name: sheep/cattle/goat

Size of area: large enough drainage; does not water logging/well drained accessibility; easy management/can get there easily closeness to connections; water/electricity; topography; flat area type of soil; stable/does not crack depth of soil; determines depth of poles/foundation; prevailing wind direction; to prevent bad smells; distance from homestead/farm house, to prevent odours. (b)

Study the income and expenses incurred by Mr Dube on a chicken

enterprise.

Bought 2 × 50 kg broiler starter mash at \$28 each
Bought 6 × 50 kg broiler finisher mash at \$25 each
Bought 100, day old chicks at \$0,75 each
Labour \$100,00
Sold 95 broilers at \$6,00 each

Calculate the farmer's gross margin.

[7]

Model Answer;

(b) Bought 2 × 50 kg broiler/starter at \$28 each = \$56,00;
Bought 6 × 50 kg broiler finisher at \$25 each = \$150,00;
Bought 100 day old chicks at \$0,75 each = \$75,00
Labour \$100 = \$100;
\$381,00
Sold 95 broilers at \$6,00 each = \$570;
Gross income 95 × 6 = \$570
Total variable costs = \$381
Gross margin = \$189

(c) Evaluate the use of vaccines in controlling viral diseases in livestock.

[5]

Question Hints and Tips

This is a higher order question that requires judgement on the use of medication that are administered in the prevention and control of diseases in livestock. Y Knowledge of vaccines and their effectiveness will help in answering the question.

You are expected to write in positive and negative aspects in the use of vaccines.

Model Answer:-

Vaccines give long term protection; cheaper; administered after a long interval; not effective after an attack; difficult to administer; need cool temperature for storage; has short live span; long withdrawal time; it is specific to an animal;

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34. (a) Discuss characteristics of an acceptable fowl run. [5]

Question Hints and Tips

This question requires knowledge of features of a suitable housing unit for broilers/indigenous chicken/layers. The learner must have studied the subtopic Small livestock Production in order to answer the question.

Model answer:-

These are the characteristics for an acceptable fowl run:-

- warmth;
- structure to provide warmth;
- rough concrete floor;
- to prevent slippery of animals;
- to prevent/protect from rain/sun;
- ventilation;
- to remove foul smells;

- (b) Outline functions of any **four** parts of the digestive system of a **named** four legged non ruminant animal. [8]

Question Hints and Tips

This question requires knowledge of the digestive system of four legged non ruminant animal which is obtained from studying sub-topic Anatomy and

physiology. The digestive system, also known as the alimentary canal is a series of organs involved in the breakdown of consumed food into smaller molecules that can be utilised by the body. It starts from the mouth and ends at the anus. Along the way are, 'accessory organs' that are needed for digestion e.g. Pancreas.

A- Non ruminant is an animal that does not chew the cud. e.g. pig, donkey or rabbit.

(b) Any four parts of the digestive system of the above mentioned ruminants would be:-

Mouth;

grinding food;

mixing food with saliva;

oesophagus;

passage way of food;

stomach;

receives/stores food;

mixes food with digestive juices;

duodenum;

chemical digestion takes place;

pancreatic juice/bile; added to

food;

to break down fats/proteins;

ileum;

absorption of food; into the blood stream;

Colon absorption of water into blood stream

Caecum contains bacteria which breaks down cellulose

Rectum stores semi-solid waste

Anus passage way for faeces

(a) Describe desirable characteristics of a **named** breeding female of a non-ruminant animal. [7]

This question requires knowledge of factors considered when breeding female of a non – ruminant animal. Breeding is mating of animals for

the production of offspring. You are required to write the names of the non-ruminants.

Model answer:-

Names of non-ruminants:- gilt/sow/jenny/doe

Characteristics

fertile;
well-developed sex
organs/teats; prolific breeder;
good mothering ability;
increased resistance to diseases;
gait/good pasture;

36. (a) State and explain two types of livestock immunity. [4]

This question requires you to narrate the types of livestock immunity and show an understanding of concepts in your own words. It requires you to give an in-depth discussion on the stated types. Immunity refers to the ability of livestock to resist disease agents to which they are exposed.

Model answer:-

Artificial immunity, acquired after an animal has been vaccinated; serum

Natural immunity; acquired after an animal has naturally recovered from a disease/ recover without being vaccinated; colostrum

- (b) Discuss the Animal Health Act of Zimbabwe. [7]

This question requires you to write about the laws governing animal health in Zimbabwe. Animal health covers animal diseases and well as the animal welfare, animal health, environmental protection and food safety.

Model answer:-

Have to detect animals incubating a disease.

Animals from red zones of notifiable infectious disease.

Should be regularly vaccinated;

Cattle should be dipped once a week in summer (Oct – April).

All animals dying from notifiable diseases must be disposed by burning/burying.

- (c) Outline management operations carried out on a calf from birth to weaning. Model. [9]

This question require you to state and give reasons for each practice.

Management operations are the day to day practices that are carried out in taking care of an animal.

A calf is a young one of a cattle,

Weaning is withdrawal of mothers' milk and gradual introduction of feeds.

Model Answer:-

Colostrum feeding; to acquire antibodies

Weighing; to get birth weigh;

Identification using ear notching/lettering/tagging/branding for record keeping

Castration; to prevent random breeding; done using open knife method/ rubber ring/burdizzo

37. (a) Describe the management of broiler chicks from **day old** to **two weeks** of age. [8]

This is question requires you to give an accurate factual account of the practices that are carried out in the care of broiler chicks.

Broilers are chickens that are kept for the production of meat.

Management practice for broiler chicks is as follows:-

Model Answer:-

Provide warmth/keeping in brooder/infrared lamp
counting;

provision of clean drinking water;

Provision of broiler starter crumps/starter mash;

Give stress pack;

Weighing;

Vaccination;

Hygiene;

biosecurity/footpath/restricted access to

brooders; protection from predators/thieves;

provision of ventilation; provision of enough space;
provision of water troughs and feed troughs
provision of medication control of diseases; adequate bedding keeping records; stress pack; vaccination; hygiene/bio security; foot bath;

- (b) Describe the life cycle of a **named** internal parasite of a farm animal.

[7]

You are required to give a detailed account on the lifecycle of a roundworm. to have knowledge of parasites that attack farm animals.

Internal parasites spend part of their lifecycle inside the gastrointestinal tract of the host including reproduction. The lifecycle can be presented diagrammatically.

The lifecycle of a round worm is as follows:-

Adult round worm in the stomach of the host animal male.

Female roundworm lays eggs

Eggs hatch to produce larvae

Larvae moults to form 2nd and 3rd stage (ineffective stage)

Grazing animal picks up the larvae

Larvae in the stomach moults into adults

Model Answer:

Name parasite - round worm in the stomach of the animal.

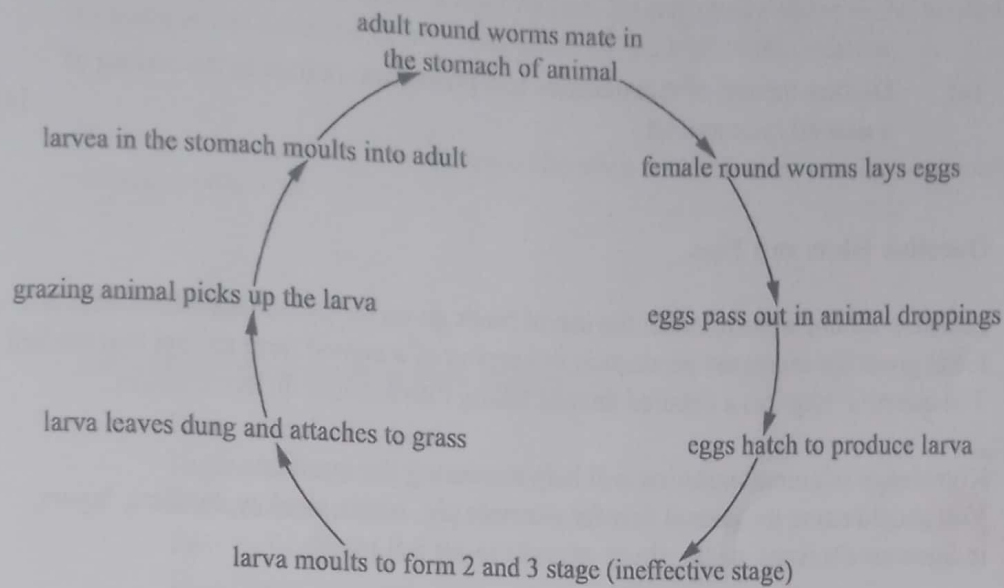
females lay eggs; eggs pass out in animal droppings; eggs hatch to produce larvae

Larva moults to form 2nd and 3rd Stage (ineffective stage)

Larva leaves dung and attaches to grass worm larva hatching out;

grazing animal picks up the larva

Lifecycle of a round worm



(c) Evaluate the importance of roughage in the diet of ruminant farm animals. [5]

Question Tips and Hints

Roughages are feeds with high fibre content per unit mass. They contain large volume and small percentage of digestible nutrients. Ruminant farm animals have four-chambered stomach; chew the cud and digest fibre. The question requires you to give the benefits of feeds high in fibre in the diet of cattle, sheep, goats.

Model answer:-

Roughage is important in the diet of ruminants because it

Adds bulk to the food/prevents constipation, contains fibre/cellulose;

Contains fibre cellulose;

Contains carbohydrates/provides energy;

Regulates water intake

Provides food for micro-organisms/fatty acids;

produce non- protein nitrogen

38. (a) Discuss the use of maintenance and production rations in the rearing of a **named** farm animal. [8]

Question Hints and Tips.

Learners should write in detail the use of feeds given for proper body functions and feeds given for increased production in keeping of a named farm animal that studied. The question requires a detailed answer taking into account different issues.

Knowledge of animal nutrition will help answering the question.

You should name the animal first for example pig, rabbit, donkey, broilers, layers, indigenous chickens, cattle, sheep or goats to get full marks.

You should indicate clearly responses for maintenance ration as well as production ration.

Model Answer:-

Name of animal

Ruminants

Cattle/sheep/goats

Non Ruminants

Pig/Rabbit/Donkey/broiler/indigenous chickens

Maintenance ration:

Food given to an animal in order to keep up body temperature.

Energy for breathing

Maintain circulation of blood;

Repair/renew worn out/damaged tissues;

Keep the animal alive;

Maintain weight/no weight gain or loss.

- (b) Discuss the advantages of using a deep litter housing system in the rearing of either a **donkey or a pig**. [7]

Question Hints and Tips

The question requires the learner to write in detail the benefits of using a system whereby new layers of straw/grass/saw dust are repeatedly added in the house when keeping either a donkey or pig. The question is a high order question.

To answer the question, knowledge on housing systems when rearing non ruminant animals is required.

Model Answer:-

Advantages of using deep litter system

Easy control of diseases;
High stocking rate;
Manure collected;
Easy to feed animals;
Less labour required;
Easy enumeration; Easy
control of parasites; Easy
to identify predators.

- (c) A farmer needs to construct a holding pen measuring $5\text{ m} \times 3\text{ m}$ with four strands of wire.

Calculate the total cost of wire needed to construct the pen, given that the wire costs \$5 per meter. [5]

Question Hints and Tips

The first part of the question requires the learner to calculate the perimeter of the pen given by the formula.

$$(L + W) \times 2$$

The perimeter give the amount of wire needed for one stand. Therefore, to obtain the total amount of wire needed you then multiply the number of strands, which is 4.

You are required to use mathematical skills on perimeter calculations. The final answer should be given with the correct units.

Answer:-

$$\text{Perimeter} = (L+W) \times 2$$

$$= (5 + 3) \times 2$$

$$= 16\text{m};$$

Total wire needed

$$= 16 \times 4 \text{ strands}$$

$$= 64\text{m}$$

$$\text{Cost is therefore} = 64\text{m} \times \$5$$

$$= \$320$$

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39. (a) Discuss the housing requirements for broilers during brooding. [8]

Question Hints and Tips

The question requires a detailed description of the housing expectations during early life of broilers where they need extra warmth. The question requires in depth discussion.

You are required to stick to the housing requirements only.

To obtain full marks, points given should be equal to or exceed the mark allocation.

Model Answer:-

Protection against heat/cold/wind/rain/bad weather;

Protection against predators;

Protection against theft;

Good ventilation;

Enough warmth;

Food, bath, biosecurity;

Bedding for warmth/comfort/absorb moisture,

Enough space,

Easy to clean;

Enough feed to troughs;
Enough water troughs;
Enough light.

- (b) Explain the management practices for either **layers** or **rabbits**, [7]

Question Hints and Tips

The learner is required to identify the animal to be taken care of, then write management practices giving reasons for each practice.

Model Answer:-

Layers

Vaccination to prevent diseases,
Feed balanced ration; for fast growth/egg production;
Supply bedding for warmth/comfort/absorption of moisture; provide clean water; for easy digestion/ for metabolic processes; clean the fowl run to prevent disease and parasites. De-beaking to reduce egg eating /reduce cannibalism;
De-worming/ dosing/drenching; to control parasites outbreak

Rabbits

Clean hutches; to prevent diseases/ parasites outbreak; vaccination to prevent diseases;
Provide a balanced ration; for fast growth;
Provide clean water; for easy digestion/ for metabolic
Nail clipping to prevent injuries /easy handling
Sexing to prevent random mating/bullying

- (c) A farmer has a deep litter system measuring **24 m × 10 m** and the

stocking rate is 12 birds/m².

- (i) Calculate the carrying capacity of the deep litter. [3]
- (ii) Assess effects of exceeding the required stocking density. [2]

Question Hints and Tips:

This question requires critical thinking and employment of mathematical skills. So to get the number of birds that can be accommodated you are required to:-

Calculate the area of the deep litter system then

Multiply by the stocking rate

Knowledge of problems associated with high population in fowl run is key in this question.

Model Answer:-

Area of the deep litter system

$$= 24\text{cm} \times 10\text{m} = 240\text{m}^2$$

$$= 240\text{m}^2 \times 12/\text{m}^2$$

$$= 2880\text{birds}$$

- (ii) Effects of exceeding the required density are
- Completion for space/food/water;
 - Poor growth rate;
 - Rapid spread of diseases;
 - Cannibalism;
 - Creates high temperatures;
 - Death.

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